

Thermodynamic State Vector Interface and Exergy Accounting in the Web of Life Ecosystem

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Sprint 014 Report

Abstract

This paper outlines the theoretical foundations and software architecture implemented in Sprint 014 of the Web of Life ecosystem. We introduce a strict thermodynamic state vector interface (`src/thermodynamics/types.ts`) and structural execution classes (`src/thermodynamics/thermodynamic`) that formalize internal entropy generation ($\dot{S}_{\text{gen}}$), exergy destruction rate ($\dot{I} = T_0 \dot{S}_{\text{gen}}$), and boundary flux arrays. By enforcing the First and Second Laws of Thermodynamics through an immutable monad pattern, the simulation guarantees rigorous energy conservation and thermodynamic consistency across all biogeochemical cycles.

1 Introduction and Systems Ecology Context

Ecosystem dynamics are fundamentally governed by thermodynamic laws. Biological organisms maintain low internal entropy at the expense of dissipating high-grade solar exergy into low-grade thermal energy. To capture these dynamics accurately, Sprint 014 establishes formal mathematical contracts within the Web of Life repository (<https://github.com/pascalranoroarijaona/WebOfLife>).

2 Thermodynamic Foundations

2.1 First Law: Conservation of Energy and Mass

For any subsystem or the global Earth Pod Ω , the internal energy change E_{sys} satisfies:

$$\frac{dE_{\text{sys}}}{dt} = \sum \dot{Q}_i - \sum \dot{W}_j + \sum \dot{m}_{\text{in}} h_{\text{in}} - \sum \dot{m}_{\text{out}} h_{\text{out}} \quad (1)$$

In our closed planetary boundary model, mass flux across boundaries is zero ($\dot{m}_{\text{net}} \approx 0$), reducing energy exchange to radiative solar inputs and longwave thermal losses.

2.2 Second Law: Entropy Generation and Exergy Destruction

The rate of entropy generation within the system is given by the Gouy-Stodola relation:

$$\dot{S}_{\text{gen}} = \frac{dS_{\text{sys}}}{dt} - \sum \frac{\dot{Q}_k}{T_k} - \sum \dot{s}_{\text{in}} \dot{m}_{\text{in}} + \sum \dot{s}_{\text{out}} \dot{m}_{\text{out}} \geq 0 \quad (2)$$

The exergy destruction rate $\dot{I}$ measures the loss of useful work potential due to irreversible processes:

$$\dot{I} = T_0 \dot{S}_{\text{gen}} \quad (3)$$

where $T_0 = 288.15$ K is the standard ambient reference temperature.

3 Software Architecture and Monadic State Transitions

To maintain transparent, immutable audit trails across simulation time steps, state transitions are encapsulated within the `ThermodynamicStateMonad`. Each state update verifies that the Second Law invariant ($\dot{S}_{\text{gen}} \geq -10^{-9}$) is strictly preserved, accounting for IEEE 754 floating-point precision limits.

4 Conclusion

Sprint 014 successfully bridges ecological systems modeling with rigorous thermodynamics, ensuring that all biogeochemical cycles within the Web of Life are physically accountable.

Official Repository: <https://github.com/pascalranoroarijaona/WebOfLife>